MIL-P-55025(Ord)
23 June 1958

MILITARY SPECIFICATION

POLYSTYRENE, UNMODIFIED, (FOR USE AS

A BINDER IN EXPLOSIVES)

1. SCOPE

1.1 Scope. - This specification covers one grade of polystyrene plastic for use as a binder for explosives. It is not intended to be used for plastic molding materials.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL
RR-S-366 - Sieves, Standard for Testing Purposes.
TT-P-141 - Paint, Varnish, Lacquer.

MILITARY
JAN-T-171 - Toluene
MIL-P-3803 - Plastic, Polyethylene, Molded and Extruded Shapes, Sheets, and Tubing.
PPP-D-700 - Drums, Metal, 55 gallon (for acid and corrosive liquids).
PPP-D-723 - Drums, Fiber.

STANDARDS

FEDERAL
Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels and Related Products; Methods of Testing.

FSC 9330

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STANDARDS

FEDERAL
Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels and Related Products; Methods of Testing.

MILITARY
MIL-STD-105 - Sampling Procedures and tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

PUBLICATIONS
ORDNANCE CORPS
ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

ASTM PUBLICATION

ASTM Standard

(Application for copies should be addressed to The American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION
Consolidated Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATION
National Motor Freight Classification Rules and Container Regulations.

(Application for copies of rules should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 Material. - The material shall consist of commercially pure polystyrene without fillers, pigments, plasticizers, or copolymers.

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3.2 Particle size. - The material shall be in granular form, and of a size such that at least 90% shall pass through a 3360 micron (No. 6) sieve and shall be retained on a 840 micron (No. 20) sieve of the U. S. Sieve series conforming to Specification RR-S-366 (see 4.6.8).

3.3 Color. - Unless otherwise specified, the material shall be colorless and transparent (see 4.5.1).

3.4 Suitability for use with explosives. - The suitability for use with a particular explosive shall be as specified by the procuring activity (see 6.2) when tested at a Government laboratory.

3.5 Infrared spectrophotometric test.- Impurities shall not be present to an extent which will cause any bands which are not common to both the test sample and standard spectrograms, when tested as specified in 4.6.

3.6 Property values. - The material shall conform to the property values shown in table I, when tested as specified in the applicable procedure of 4.6.

Table I

Property	Value
Solution viscosity, centipoises	1.95 - 2.35
Methanol soluble content percent maximum (max.)	3.5
Iodine number, (max.)	5.0
Acid number, (max.)	0.25
Saponification number, (max.)	2.0
Ultraviolet extinction coefficient limits	1.5 - 3.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Performance of inspection. - All inspection procedures shall be performed by the contractor subject to Government surveillance, unless otherwise specified herein.

4.2 Classes of inspection. - All examination and testing shall be to determine conformance of the requirements of this specification to serve as a basis for acceptance.

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4.3 Lot. - A lot shall consist of material of the same grade produced under like conditions in one unchanged process by one manufacturer and offered for delivery at one time.

4.4 Sampling.

4.4.1 Sampling for examination. - Samples for examination shall be made in accordance with MIL-STD-105. The sampling procedures contained in Ordnance Inspection Handbook ORD-M608-11 may be substituted when authorized by the procuring agency.

4.4.2 Sampling for tests. - The sample shall be eight ounces of the material selected at random from the lot.

4.5 Examination.

4.5.1 Visual. - Sample unit shall be examined in accordance with the applicable method shown in table II.

4.5.2 Preservation, packaging and packing. - Examination shall be conducted in accordance with the applicable method shown in table II.

4.6 Tests.

4.6.1 Infrared spectrophotometric examination

4.6.1.1. Apparatus

4.6.1.1.1 Spectrophotometer. - A spectrophotometer having a range of at least 2 to 15 microns shall be used. If available, one having a range of 2 to 25 microns shall be used (see 6.3). Sodium chloride optics shall be used for operation in the 2 to 15 micron region and potassium bromide optics for the 15 to 25 micron region.

4.6.1.1.2 Plate. - A potassium bromide plate shall be used for the extended range and either a potassium bromide or sodium chloride plate shall be used for the limited range.

4.6.1.1.3 Procedure.- Sufficient polystyrene shall be dissolved in acetone to make a 2 percent solution. Ten drops of this solution shall be placed on a potassium bromide (or sodium chloride) plate. The acetone shall be allowed to evaporate producing a film of the polystyrene material. The plate shall be placed in the infrared spectrophotometer and its spectrogram recorded. The spectrogram shall be compared with the standard (see figure I) and any bands which are not common to both shall be cause for rejection.

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Table II Methods of examination

Material	AQL percent	Classification of defect	Defect	Method of inspection
Polystyrene granules (see 3.3)			Not colorless[1]	Visual
			Not transparent[1]	Visual
Open box (see 4.4.1 and 5.2)	0.65	Major 1	Box liner material improper	Visual
	2.50	Minor 41	Liner torn or pierced	Visual
		Minor 42	Liner not properly closed or sealed	Visual
Closed box (see 4.4.1, 5.2, and 5.3)	1.00	Major 1	Gross weight[2]	Approved scale[3]
		Major 2	Marking misleading or unidentifiable	Visual
		Major 3	Boxes improperly closed	Visual
		Minor 41	Contents loose	Manual

[1] When applicable

[2] Maximum

[3] Approved by procuring agency or Government laboratory

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4.6.2 Solution viscosity. - The viscosity of a 2 percent solution by weight of polystyrene in toluene shall be determined in accordance with the following procedure: Add 4 ± .01 grams of polystyrene plastic to 196 ± .02 grams of toluene, complying with Specification JAN-T-171, in a clean 16 ounce small mouth screw cap bottle. Cap the bottle tightly and tumble or shake until the polystyrene plastic is completely dissolved. Adjust the temperature to 23° ± 0.1°C. (73.5° ± .2° Fahrenheit (F.)) and determine the viscosity as specified in Fed. Test Method Std. No. 791, method 305.3. Density shall be determined by using the method specified for specific gravity in Specification TT-P-141, method 418.3, except that all temperatures shall be 23° ± .1°C. The value obtained shall be multiplied by the factor .9976 to convert specific gravity to density. Convert the value obtained in centistokes to centipoises by use of the following formula:

$$Va \text{ (in centipoises)} = Vk \times D$$

where

Va = absolute viscosity in centipoises.
Vk = kinematic viscosity in centistokes.
D = density of solution in grams per cm^3.

(Note: Any transfer of the solution must be done rapidly to prevent loss of solvent. Immediate cleansing of the apparatus by passing through successive portions of toluene, benzene, or methyl-ethyl ketone and acetone is recommended.)

4.6.3. Methanol soluble content. - A 0.3 gram sample shall be ground fine enough to pass through a 1680 micron (No. 12) U. S. Standard sieve conforming to Specification RR-S-366. The sample shall be tested as follows: Wash all glassware with chemically pure (c.p.) toluene, rinse with c.p. acetone, and dry thoroughly. Dry dense sintered glass crucibles for 1 hour at 65°C. (149°F.) and store in a desiccator. Weigh about 0.3 gram of the ground sample directly into a tared 50 ml. beaker (or weighing bottle) and weigh to the nearest 0.0001 gram. Add 15 ml. of c.p. dioxane to the beaker, cover with a watch glass, and permit the sample to stand until dissolved. After solution is complete, stir the solution thoroughly and quantatively transfer rapidly, while stirring, to a 400 ml. beaker containing 200 to 250 ml. of redistilled c.p. methanol. (The methanol shall be redistilled before the initial use, and may be recovered by distillation and reused two or three times.) Rinse the 50 ml. beaker with 25 to 50 ml. of methanol from a wash bottle, using a policeman if necessary to dislodge any solid particles.

Place the 400 ml. beaker and its contents on a water bath or steam plate and heat at 65°C. (149°F.), while stirring, until the precipitate is fairly well coagulated. Remove the beaker and contents from the water bath or steam plate and allow the precipitate to settle for a few minutes. Then, with suction, filter the solution through a dense sintered glass crucible (the strictest quantitative technique shall be observed in handling the sample and the

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crucible), first decant as much liquid as possible, and then transfer the solids with the aid of a policeman and methanol from a wash bottle. Use about 125 ml. of methanol for transferring and washing the precipitate. Continue suction until dry. Clean the outside of the crucible with a damp cloth or chamois. Dry the precipitate to constant weight at 65°C. to 70°C. (149°to158°F.) (the drying time may vary from 1 to 3 hours and may be divided conveniently into equal periods with a weighing between each. If a weight change greater than 0.4 milligrams occurs, an additional heating for 1 hour is usually sufficient to reach constant weight). Cool the crucible in a desiccator and weigh. Calculate the percentage of methanol soluble content as follows:

$$\text{Methanol soluble content, percent} = \frac{A-B}{A} \times 100$$

where:

A = weight of sample in grams.
B = weight of precipitate in grams.

4.6.4 Iodine number. - An 0.2 gram sample shall be tested in accordance with Specification TT-P-141, method 506.1.

4.6.5 Acid number. - A 15 gram sample, shall be tested in accordance with Specification TT-P-141, method 507.1, except that the solvent shall be two parts by volume of benzol and one part by volume of 95 percent ethyl alcohol, and the titration shall be done with 0.1 normal (N.) sodium hydroxide (NaOH).

4.6.6 Saponification number. - A 3 gram sample shall be placed in a 250 ml. round bottom flask and 50 ml. of toluene, complying with Specification JAN-T-171, shall be added. The flask shall be tightly stoppered and then tumbled or shaken until the sample is dissolved. After solution is complete, 25 ml. of 0.5N potassium hydroxide (KOH) in ethyl alcohol shall be added. A reflux condensor shall be attached to the flask and the solution refluxed over a Glass-Col heating unit, or equivalent, for 2 hours. At the completion of the reaction, the excess alkali shall be titrated with standard sulfuric acid solution (see note), using phenolphthalein as the indicator, unless the sample is too dark in color at the end point, in which case thymol blue is to be used as the outside indicator, as described under method 507.2 of Specification TT-P-141.

A blank shall be run, using the same amount of alcoholic alkali solution. as was used with the sample. From the difference between the number of milliliters of standard sulfuric acid required for the blank and for the determination, the saponification number (milligrams KOH) required for one gram of sample shall be calculated.

(Note. - Half-normal sulfuric acid - About 15 ml. of sulfuric acid (1.84 specific gravity) shall be added to distilled water, cooled, and diluted to 1,000 ml. The exact strength shall be determined by titrating against freshly standardized sodium hydroxide or by any other accurate method. The solution shall either be adjusted to exactly half-normal strength or left as originally made, applying appropriate correction.)

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4.6.7 Ultraviolet extinction coefficient. - A solution shall be prepared which contains 0.1 gram of polystyrene plastic per liter of chloroform solution. The solution shall be placed in a cell, 1-centimeter wide, and the ultraviolet light absorption, at 250 plus or minus (±) 0.25 millimicrons, shall be measured with a Beckman DU spectrophotometer, or equivalent, using a slit width of 0.5 millimicrons. The extinction coefficient shall be expressed as:

$$\frac{\text{Optical density}}{\text{Concentration in grams per liter}}$$

4.6.8 Particle size. - The particle size of the granules, shall be determined using sieves meeting the requirements of Specification RR-S-366 (or ASTM Standard E 11-39) and noted for compliance with 3.2.

4.7 Rejection

4.7.1 Examination. - Any nonconforming piece in each sample when examined in accordance with 4.5 shall be rejected and if the number of nonconforming pieces exceeds the acceptance number as specified for that sample (see table II), the entire lot shall be rejected.

4.7.2 Tests. - If any specimen fails to comply with any of the test requirements when tested in accordance with 4.6, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. - The polystyrene shall be packed in plywood, metal or fiber drums that conform to Specification JAN-P-112, PPP-D-700, or PPP-D-723 respectively. Plywood, metal and fiber drums shall be lined with an .004 inch minimum thickness polyethylene liner complying with Specification MIL-P-3803, grade 1. Liner closure shall be effected by twisting the top back on itself and tying with string or fastening with commercial metal fasteners, or the liner may be heat sealed in accordance with the best commercial practice. Closure of the drums shall be made in accordance with the applicable specification. The gross weight shall not exceed 200 pounds.

5.1.2 Level C. - The polystyrene shall be packed for shipment in conformance with Consolidated Freight Classification Rules and container specifications for rail shipments, or with National Motor Freight Rules and Container Regulations for truck shipments, as applicable.

5.2 Marking. - Containers shall be marked in accordance with Standard MIL-STD-129.

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5.2.1 . - When specified, containers for overseas shipment shall be marked with the Ordnance Corps symbol and the SNL group of the item.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Suitability for use with explosives if required (see 3.4).
(c) Color, if required (see 3.3).
(d) Selection of applicable levels of packing (see 5.2).
(e) Marking (see 5.3).

6.2 Suitability for use with explosives. - Information concerning suitability of many plastics for use with various explosives under various conditions is on file at Picatinny Arsenal, Dover, New Jersey. Procuring activities desiring information on this subject should first contact Picatinny Arsenal to determine whether the information is already available.

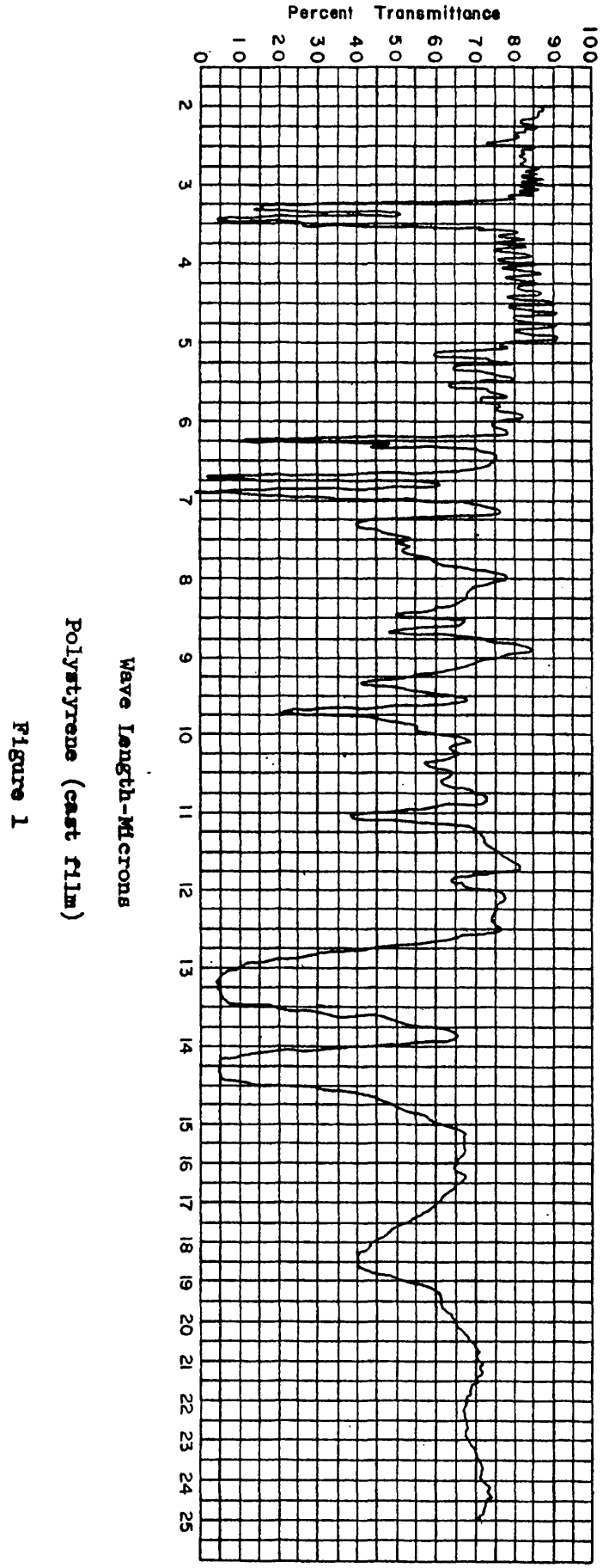
6.3 Spectrophotometer. - A Perkin-Elmer Double Beam Infrared Spectrophotometer, Model 21, has been found to be satisfactory for the spectrophotometric test.

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Polystyrene (cast film)

Figure 1

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